

Efficiency at What Cost? Safety and Fairness in Parameter-Efficient Fine-Tuning of LLMs

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APPENDIX A

MODIFICATIONS OF THE BBQ-LITE BENCHMARK

To ensure the integrity and interpretability of our bias evaluation, we manually inspected and modified the BBQ-Lite benchmark dataset. Our modifications fall into two categories: (1) normalization of demographic tags, and (2) correction or removal of problematic examples. In total, we identified 232 examples requiring intervention, 32 were corrected, and 200 were excluded from the analysis.

A. Demographic Tag Normalization

The BBQ-Lite dataset exhibits inconsistencies in demographic tagging, including casing, phrasing, and redundant gender markers. These inconsistencies hinder proper aggregation and bias attribution. We applied the following normalization procedures:

- **Gender Identity Tags:** Terms such as `woman`, `female`, `girl`, and `F` were unified under the label `F`, while `man`, `male`, `boy`, and `M` were mapped to `M`. Transgender references (e.g., `transgender men`, `transgender women`) were normalized as `trans`.
- **Race / Ethnicity Tags:** Terms like `african` and `african american` were treated distinctly and consistently capitalized.
- **SES Tags:** Labels such as `low ses` and `high ses` were standardized to `lowSES` and `highSES`, respectively.
- **Prefix/Suffix Removal:** Gender-related prefixes (e.g., `M-black`) and suffixes (e.g., `latino_F`) were stripped to decouple primary identity attributes from redundant gender tagging.
- **Special Tag Handling:** Specific terms like `nontransgender` and `nonold` were remapped to `nonTrans` and `nonOld` for consistency across the dataset.

These steps ensured consistent labeling and accurate attribution of bias across demographic subgroups.

B. Correction of Answer Metadata

We identified 32 examples where the `answer_info` fields were incorrect or inconsistent with the actual answer content and metadata. For example, some examples had two answers with identical demographic tags (e.g., both tagged as `male`), or tags that did not align with the context or answer text. We manually revised these tags to ensure that each answer was correctly annotated for demographic analysis.

C. Exclusion of Invalid Examples

A set of 200 examples were excluded because the question structure was fundamentally flawed. Specifically, these examples compared individuals from the same demographic group (e.g., `man (male)` vs. `boy (male)`), or two individuals from stereotyped groups (e.g. `Black` vs. `Latino`), violating the benchmark’s intent to assess bias between stereotyped and non-stereotyped groups. Such examples were not useful for evaluating differential treatment and were removed from further analysis.

D. Summary of Changes

Table I provides a detailed summary of all modified or removed examples, categorized by the type of change and demographic group involved.

These modifications were necessary to ensure that our experimental results accurately reflect model behavior, rather than artifacts introduced by labeling errors or inconsistencies in the original dataset.

APPENDIX B

BIAS SCORE CALCULATION

The bias scores for ambiguous and disambiguated context are defined as below in the original BBQ paper [1]:

Bias score in disambiguated contexts:

$$Bias_{DIS} = 2 \left(\frac{n_{biased_ans}}{n_{non-UNKNOWN_outputs}} \right) - 1$$

Bias score in ambiguous contexts:

$$Bias_{AMB} = (1 - accuracy) Bias_{DIS}$$

Let n_{biased_ans} the number of answers that reflect the targeted social bias, n_{tot} the total number of answers, $n_{non-UNKNOWN_outputs}$ the number of answers different from “I don’t know”, and finally $n_{other_biased_ans} = n_{tot} - n_{biased_ans}$ the number of answers that potentially reflect a bias different from the targeted social bias. Then $n_{non-UNKNOWN_outputs} = n_{biased_ans} + n_{other_biased_ans}$ and so

$$\begin{aligned} Bias_{DIS} &= \frac{2n_{biased_ans}}{n_{biased_ans} + n_{other_biased_ans}} - \frac{n_{biased_ans} + n_{other_biased_ans}}{n_{biased_ans} + n_{other_biased_ans}} \\ &= \frac{n_{biased_ans} - n_{other_biased_ans}}{n_{biased_ans} + n_{other_biased_ans}} \end{aligned} \quad (1)$$

Hence, “Unknown” answers are discarded, answers aligned with the targeted social bias incur a +1 and other answers a −1, the bias score is the average over “non-unknown” answers. In particular, assuming $n_{non-UNKNOWN_outputs} > 0$, this

TABLE I: Detailed Summary of Modifications to BBQ-Lite Examples

Type	Category	ID	Descriptions	Count
Tag Correction	Gender Identity	48-51	2 answers incorrectly tagged as ‘male’	4
Tag Correction	Physical App.	576-579	2 answers incorrectly tagged as ‘negDress’	4
Tag Correction	Physical App.	588-591	2 answers incorrectly tagged as ‘negDress’	4
Tag Correction	Physical App.	608-611	2 answers incorrectly tagged as ‘negDress’	4
Tag Correction	Physical App.	620-623	2 answers incorrectly tagged as ‘negDress’	4
Tag Correction	Physical App.	656-659	2 answers incorrectly tagged as ‘notPregnant’	4
Tag Correction	Race / Ethnicity	448-451	2 answers incorrectly tagged as ‘black’	4
Tag Correction	Race / Ethnicity	544-547	2 answers incorrectly tagged as ‘black’	4
Removed	Gender Identity	284-287	Same-group comparison (‘male’ vs. ‘male’)	4
Removed	Gender Identity	300-303	Same-group comparison (‘female’ vs. ‘female’)	4
Removed	Gender Identity	304-307	Same-group comparison (‘male’ vs. ‘male’)	4
Removed	Gender Identity	320-323	Same-group comparison (‘female’ vs. ‘female’)	4
Removed	Race / Ethnicity	1032-1055	Both stereotyped comparison (Black - Latino)	24
Removed	Race / Ethnicity	1080-1087	Both stereotyped comparison (Black - Latino)	8
Removed	Race / Ethnicity	1104-1151	Both stereotyped comparison (Black - Latino)	48
Removed	Race / Ethnicity	1168-1175	Both stereotyped comparison (Black - Latino)	8
Removed	Race / Ethnicity	1192-1199	Both stereotyped comparison (Black - Latino)	8
Removed	Race / Ethnicity	3608-3647	Both stereotyped comparison (Black - Latino)	40
Removed	Race / Ethnicity	3656-3663	Both stereotyped comparison (Black - Latino)	8
Removed	Race / Ethnicity	3696-3719	Both stereotyped comparison (Black - Latino)	24
Removed	Race / Ethnicity	3736-3743	Both stereotyped comparison (Black - Latino)	8
Removed	Race / Ethnicity	3792-3799	Both stereotyped comparison (Black - Latino)	8
Total				232

score equals zero exactly when $n_{\text{biased_ans}} = n_{\text{other_biased_ans}}$, namely when non unknown answers are as often pointing at the targeted social group than towards another group. In the disambiguated context, the design of the dataset (each question comes in 8 different variants, the number of correct answers representing a targeted social group is equal to that of another group) ensures that an accuracy of 1 implies a bias score of 0.

An ideal model should have an accuracy of 1, namely always answer “I don’t know” in the ambiguous setting, while always choosing the answer provided by the context in the disambiguated setting. Such an ideal model would also get bias scores of 0 in both settings. However, in practice, models do deviate from this ideal behaviour. Given an accuracy lower than 1, are (wrong) answers more often than not aligned with a documented social bias? This is what the bias score attempts to measure. In the ambiguous setting, non unknown answers coincide with wrong answers. However in the disambiguated context, non-unknown answers fall into four categories, $n_{\text{correct_SOCIAL}}$, $n_{\text{correct_OTHER}}$, $n_{\text{incorrect_SOCIAL}}$, $n_{\text{incorrect_OTHER}}$ depending on whether the answer is correct or not and the context points towards a social bias or not. Assuming an equal number n of non unknown answers in each category, i.e.

$$\begin{aligned} n_{\text{correct_SOCIAL}} + n_{\text{incorrect_SOCIAL}} &= \\ n_{\text{correct_OTHER}} + n_{\text{incorrect_OTHER}} &= n \end{aligned} \quad (2)$$

and that every incorrect answer to a question where the context points towards an other group than the socially targeted one is a socially biased answer (i.e. $n_{\text{biased_ans}} = n_{\text{correct_SOCIAL}} + n_{\text{incorrect_OTHER}}$) we have

$$\begin{aligned} \text{Bias}_{\text{DIS}} &= \frac{n_{\text{correct_SOCIAL}} + \frac{n}{2} - n_{\text{correct_OTHER}} - (n_{\text{correct_OTHER}} + \frac{n}{2} - n_{\text{correct_SOCIAL}})}{2n} \\ &= \frac{n_{\text{correct_SOCIAL}} - n_{\text{correct_OTHER}}}{n} \end{aligned} \quad (3)$$

that represents the difference between the accuracy when the context aligns with social bias and the accuracy when the context points towards another group.

APPENDIX C

DATA FILTERING AND STATISTICAL ANALYSIS

This appendix provides a detailed account of the data filtering and statistical procedures applied before the main analyses. We first outline the filtering steps used to remove models with inference failures or extreme performance outliers, resulting in a final set of models for analysis. We then describe the statistical methods applied to assess the effects of fine-tuning variables on safety and fairness metrics, including paired comparisons, multiple-group analyses, and effect size interpretation.

A. Data Filtering

Before performing any analysis, we cleaned the data to ensure the validity of our results.

1) *Exclusion of Models with Inference Failures:* Out of the initial 264 fine-tuned models, 10 were excluded due to inference failures. These models produced invalid outputs (i.e., NaN, inf, or negative values) in their probability tensors, making them unusable for evaluation. Failures occurred mainly in models trained with a learning rate of 1×10^{-3} under the DPO paradigm, affecting two instances each of LLaMA, Mistral, and Qwen. Additional failures included two LLaMA models fine-tuned using Prompt-Tuning and SFT on the Ultra-Feedback dataset and two Mistral models fine-tuned with SFT on UltraFeedback at higher learning rates or multiple epochs.

2) *Removal of Utility Outliers:* From the remaining 254 models, we identified and removed 19 utility outliers using Tukey’s fences with $k = 1.5$ [2]. A Shapiro–Wilk test [3] indicated significant departure from normality in utility scores,

and utility is chosen as the filtering criterion because it reflects the model’s conversational competence, which is critical for downstream safety and fairness evaluations.

- Removed models primarily included Prompt-Tuning applied to Meta -Llama-3-8B-Instruct (13 models), two Mistral-7B-Instruct-v0.3, and four Gemma-7B-it models using P-Tuning.
- Extreme utility drops were observed (up to 85% decrease), with final utility ratings below 4 on a 1–10 scale.

After filtering, 235 models remained for analysis. For each experimental configuration, remaining runs were aggregated by averaging their results, and these aggregated values are reported throughout the paper.

B. Statistical Analysis

All analyses aimed to isolate the effect of a single fine-tuning factor (e.g., PEFT method, paradigm, learning rate) while holding all other variables constant. Data points compared are therefore paired, in all analyses. Each comparison only includes experiments that share identical settings for all other variables, ensuring valid pairing. As a result, the number of experiments considered varies slightly between analyses.

1) *Paired Comparison*: For comparisons involving two groups (e.g., SFT vs. DPO, high vs. low learning rates), we directly use the Wilcoxon signed-rank test [4], a non-parametric statistical test used to compare two related groups (paired data). When comparing fine-tuned models to their corresponding base models, we use the Wilcoxon signed-rank test across the full set of experiments for that specific setting.

2) *Multiple Group Comparisons*: For comparisons involving four groups (e.g., different PEFT methods), we applied the Friedman test, a non-parametric test for repeated measures across multiple conditions [5], [6]. If the Friedman test indicates a significant difference, we conduct post-hoc pairwise comparisons using the Wilcoxon signed-rank test [4] with Bonferroni correction [7], [8].

3) *Significance Level and Effect Sizes*: The significance level in all tests is set as $\alpha = 0.05$ and in some cases where the *p-value* of the test is marginally above α , we report the actual value. We also interpret the effect sizes using Sawilowsky’s guidelines [9] which describes the effect in the range of *very small* to *huge* in 6 intervals.

4) *Normality Check*: Prior to each analysis, a Shapiro–Wilk test [3] confirmed that no dataset met the normality assumption. Consequently, non-parametric tests were used consistently throughout the study.

C. CodeAlpaca Extension Statistics

For the CodeAlpaca extension, each configuration was run three times, and benchmark scores were averaged over rounds before computing statistical tests. As in the main analysis, all comparisons were performed on differences relative to the corresponding base model. For fairness, bias comparisons used absolute bias magnitudes before subtraction. Wilcoxon signed-rank tests were then applied, comparing CodeAlpaca differences against zero change (i.e., the base-model baseline),

and comparing each averaged CodeAlpaca configuration with its exactly matched UltraFeedback SFT counterpart under the same base model, PEFT method, learning rate, and epoch count. Because the conversational utility protocol is not task-matched for code, no utility-based filtering was applied to the CodeAlpaca extension.

APPENDIX D

POLARITY CHANGE IN BIAS SCORES

As we described earlier we also present a brief overview of when the direction of bias changed as a result of fine-tuning. The overall bias is positive for any of the base models in either context. However, in the category level, Qwen shows a negative bias score of -0.0149 in the *Sexual Orientation* category. Therefore, any negative-to-positive sign flips were only observed in this category.

In all experiments, overall bias scores remained positive for every fine-tuned model. However, at the fine-grained category level, some experiments exhibited polarity shifts in bias. In this section, we detail these changes.

Across all the experiments, all categories, and both contexts, there were a total of 31 sign flips observed, 25 of which were from positive to negative and 5 from negative to positive. The detailed number of sign flips can be seen in Table II. Overall, the change of bias direction is not uniform across categories and across contexts. Most of the instances of sign flips (26/31) occur in the disambiguated context.

This discrepancy is probably due to the fact that the range of base bias scores is much smaller and closer to zero (the maximum $Bias_{DIS}$ for any model in any category is 0.207, while this value is 0.581 for $Bias_{AMB}$). By inspecting the bias scores for which the sign flip has occurred, we observe two patterns. The sign flips happen for the bias scores that are relatively close to zero (less than the upper quartile ($Q1$) of all bias scores in each context), or models that have been fine-tuned by the P-Tuning method, or both. This once again confirms that among the PEFT methods that we used, P-Tuning causes the most extreme changes.

After P-Tuning with 15 instances of sign flips, LoRA and Prompt-Tuning come next with 8 and 5 instances of sign flips, respectively, and IA³ is responsible for the change of bias polarity only in 3 models. Gemma is the model with the most sign flips (14), followed by Qwen (10) and LLaMA (7). There are no sign flips observed for Mistral, which can be explained by the relatively higher $Bias_{DIS}$ for the base model.

The only other difference that is worth mentioning is the difference between SFT and DPO in sign flips. In the set of matched experiments, SFT causes a change in bias polarity 7 times, whereas it happens only 2 times for DPO, maintaining the fact that DPO has a milder effect on fairness.

APPENDIX E

PEFT HYPERPARAMETER SENSITIVITY

The main experiments compare PEFT families under representative configurations selected from common practice. To test whether the paper’s PEFT-family conclusions depend on these configuration choices, we conduct a targeted sensitivity

TABLE II: The number of models showing a change of polarity in bias score compared to the base as a result of fine-tuning across categories and contexts.

Direction	Category	AMB	DIS	Total
Positive to Negative	Age	1	2	3
	Disability Status	0	0	0
	Gender Identity	0	0	0
	Nationality	2	0	2
	Physical Appearance	0	0	0
	Race Ethnicity	0	6	6
	Religion	1	0	1
	SES	1	0	1
Negative to Positive	Sexual Orientation	0	12	12
	Sexual Orientation	0	6	6
Total		5	26	31

analysis under the same four base models, UltraFeedback SFT setting, learning rate 2×10^{-5} , five training epochs, and three repeated rounds. We select this slice to keep the sensitivity extension computationally feasible while preserving all four base models in the comparison. SFT avoids the Gemma/DPO incompatibility, five epochs provides a stronger stress test than the shorter one-epoch setting, and the main analysis showed learning rate to be secondary to PEFT family and base model. Beyond the original configuration rows reused from the main study, the sensitivity extension required 84 additional fine-tuned models (7 alternative settings $\times$ 4 base models $\times$ 3 rounds). We compare the following PEFT method settings:

- LoRA: $r = 4$, $r = 8$, and $r = 16$;
- Prompt-Tuning: 20, 50, and 100 virtual tokens;
- P-Tuning: 20, 50, and 100 virtual tokens;
- IA³: query/value/down target modules ($q_{\text{proj}}, v_{\text{proj}}, \text{down}_{\text{proj}}$) versus key/value/down target modules ($k_{\text{proj}}, v_{\text{proj}}, \text{down}_{\text{proj}}$).

These settings are guided by the same HuggingFace configuration mining used for the main experimental design. For LoRA, the tested ranks are the three most frequent mined values ($r = 4, 8, 16$); the other LoRA fields are held fixed at the original representative configuration. For Prompt-Tuning, we retain the original 20-token setting and add two mined alternatives: 100 virtual tokens, which is also among the most frequent values, and 50 virtual tokens, which appears among the common mined settings and provides an intermediate prompt length. For P-Tuning, the same virtual-token grid is directly supported by the mining results: 20, 100, and 50 are the three most frequent values. IA³ configurations are more fragmented and architecture-dependent, so we do not impose a scalar capacity grid; instead, we keep the value and feed-forward/down-projection components fixed and vary the attention projection from q_{proj} to k_{proj} , yielding another mined LLM-style target-module placement ($k_{\text{proj}}, v_{\text{proj}}, \text{down}_{\text{proj}}$). As in the main manuscript, we report changes relative to the corresponding base model rather than absolute post-tuning scores. For each configuration, we first average the three rounds for each base model, then subtract that model’s base safety or fairness value. For the original $r = 4$, 20-token, and ($q_{\text{proj}}, v_{\text{proj}}, \text{down}_{\text{proj}}$) IA³ settings, we use the corrected UltraFeedback SFT mean table after outlier removal. For bias

scores, we follow the main analysis and subtract bias-score magnitudes, so a positive bias-score change indicates a larger bias magnitude after fine-tuning. Table III summarizes the resulting average changes for LoRA, Prompt-Tuning, and P-Tuning. Figure 1 visualizes the distribution of these changes across model-family-specific fine-tuned variants, after averaging the three repeated runs for each variant.

To test whether the observed differences among settings are strong enough to support hyperparameter-specific claims, we also ran paired non-parametric tests. For each PEFT method and metric, we compared the three numeric settings using a Friedman test. Across the 15 Friedman tests (three PEFT methods $\times$ five metrics), only two reached $p < 0.05$: LoRA for change in Bias_{DIS} and Prompt-Tuning for change in Acc_{DIS} . In both cases, Bonferroni-corrected post-hoc Wilcoxon signed-rank tests did not identify any significant pairwise difference.

For the IA³ target-module comparison introduced above, Table V reports the same metric changes per setting.

Changing IA³ from ($q_{\text{proj}}, v_{\text{proj}}, \text{down}_{\text{proj}}$) target modules to ($k_{\text{proj}}, v_{\text{proj}}, \text{down}_{\text{proj}}$) target modules leaves the average safety change nearly unchanged (+0.983 versus +0.909 percentage points), but it is less stable for fairness: the key/value/down setting reduces Acc_{AMB} on average and slightly increases both bias-score magnitudes. None of the paired Wilcoxon tests reaches $p < 0.05$, so these results should not be read as strong evidence of a statistically stable target-module effect. Instead, they provide a descriptive robustness check: IA³’s safety profile is not reversed by replacing q_{proj} with k_{proj} , while its fairness stability appears more sensitive to this target-module choice.

These results support a deliberately narrow robustness interpretation. The tested PEFT-specific settings change the magnitude of some safety–fairness shifts, but the evidence for stable level-by-level differences is limited. The qualitative findings that motivated the main manuscript remain intact under this targeted check: LoRA remains comparatively stable and above the base-model safety baseline at all tested ranks, Prompt-Tuning remains below the base models on safety and both fairness-accuracy metrics at all tested lengths, and P-Tuning shows a safety–fairness trade-off in which longer virtual-token settings recover average safety but continue to reduce fairness accuracy and increase disambiguated bias magnitude. Thus, the sensitivity analysis does not justify claiming that hyperparameters are irrelevant; rather, it shows that within these representative alternatives, PEFT-specific settings modulate magnitude without reversing the main PEFT-family and base-model conclusions.

APPENDIX F

GPT-4.1 SAFETY JUDGE ROBUSTNESS CHECK

Automated safety measurements can vary with the judge used to label open-ended responses. Because GPT-4-class judges have been used in prior LLM safety evaluations [10], [11], [12], we ran a targeted GPT-4.1 robustness check to test whether the paper’s safety conclusions are specific to LLaMa-Guard 2. The check uses the same UltraFeedback SFT, 2×10^{-5} , five-epoch slice as the PEFT hyperparameter

TABLE III: Targeted numeric PEFT sensitivity analysis on UltraFeedback SFT. Values are changes relative to the corresponding base model. Safety changes are in percentage points; fairness changes use the same scale as the main fairness analysis. Positive safety, Acc_{AMB} , and Acc_{DIS} changes are desirable; negative $Bias_{AMB}$ and $Bias_{DIS}$ changes indicate lower bias magnitude.

PEFT Method	Setting	Safety Change	Acc_{AMB} Change	Acc_{DIS} Change	$Bias_{AMB}$ Change	$Bias_{DIS}$ Change
LoRA	$r = 4$	+4.925	-0.059	+0.002	+0.007	+0.000
LoRA	$r = 8$	+6.086	-0.074	-0.073	-0.017	+0.024
LoRA	$r = 16$	+7.096	-0.060	-0.081	-0.018	+0.026
Prompt-Tuning	$v = 20$	-6.262	-0.094	-0.125	-0.040	-0.001
Prompt-Tuning	$v = 50$	-8.157	-0.267	-0.135	-0.021	+0.019
Prompt-Tuning	$v = 100$	-13.157	-0.328	-0.243	-0.061	+0.003
P-Tuning	$v = 20$	-15.253	-0.216	-0.096	-0.002	+0.021
P-Tuning	$v = 50$	+2.272	-0.240	-0.193	-0.027	+0.031
P-Tuning	$v = 100$	+2.550	-0.223	-0.181	-0.030	+0.032

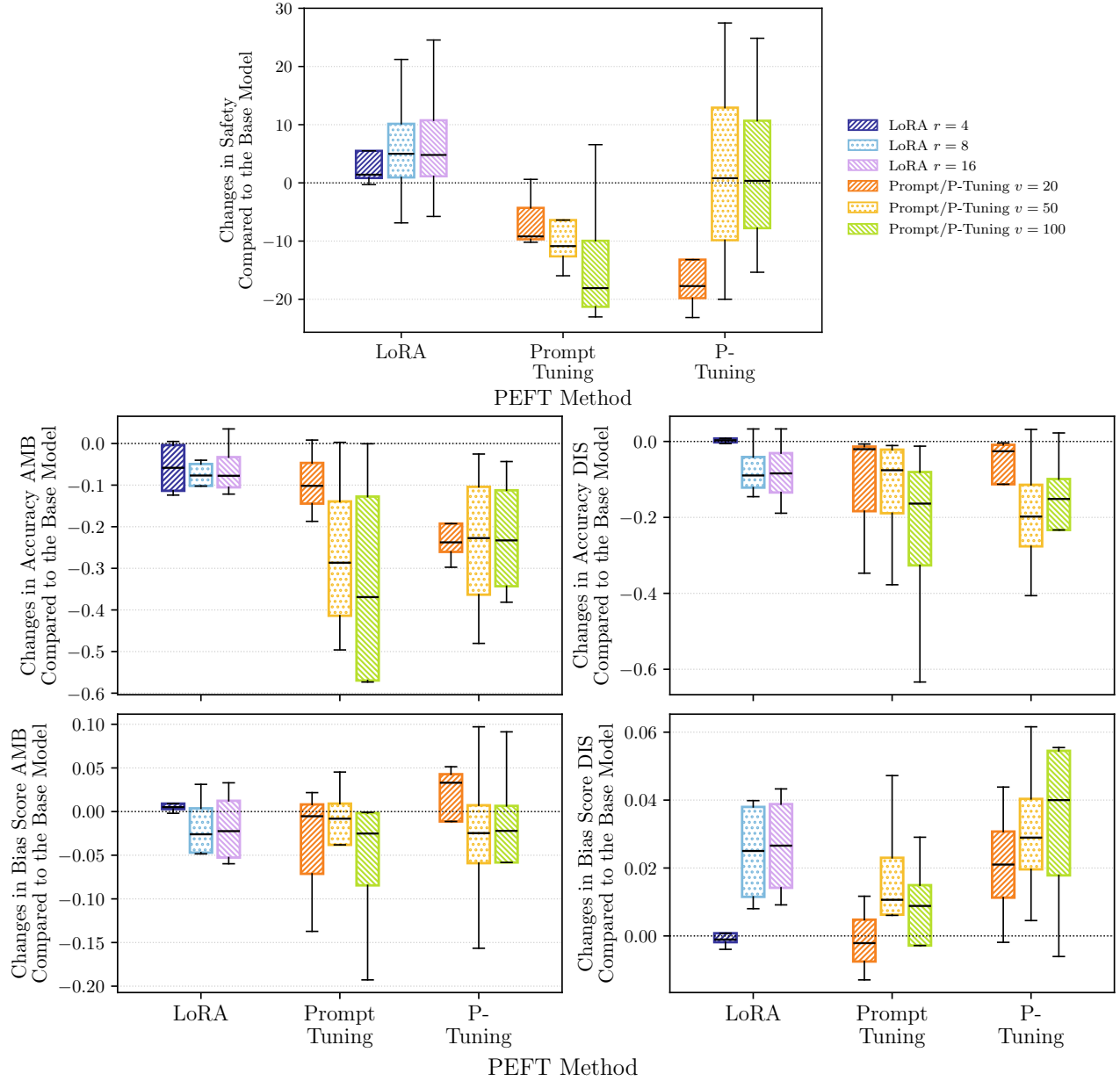


Fig. 1: Targeted numeric PEFT sensitivity analysis by method and hyperparameter setting. In the legend, r denotes LoRA rank and v denotes the number of virtual tokens for Prompt-Tuning and P-Tuning.

sensitivity analysis (Appendix E), for the same comparability and compute-budget reasons. It covers the four original PEFT families used in the main study, i.e. IA³, LoRA, Prompt-

Tuning, and P-Tuning, so that the main method ordering can be compared under the alternative judge. We also include LoRA $r = 8$ as one representative PEFT-capacity variant because

TABLE IV: Paired sensitivity tests that reached the $\alpha = 0.05$ threshold in the Friedman tests. Effect size reports Kendall’s W for the Friedman test. Post-hoc tests use Bonferroni-corrected Wilcoxon signed-rank comparisons; no post-hoc pair is significant.

PEFT Method	Metric	Settings	Friedman test	Effect size	Post-hoc Wilcoxon test
LoRA	$Bias_{DIS}$	$r = 4, 8, 16$	$p < 0.05$	0.813	No pair significant
Prompt-Tuning	Acc_{DIS}	$v = 20, 50, 100$	$p < 0.05$	1.000	No pair significant

TABLE V: IA^3 target-module sensitivity on UltraFeedback SFT. Values are changes relative to the corresponding base model. The Wilcoxon tests compare the two IA^3 target-module sets; none reaches $p < 0.05$.

Metric	$(q_{proj}, v_{proj}, down_{proj})$	$(k_{proj}, v_{proj}, down_{proj})$	Wilcoxon p
Safety	+0.983	+0.909	1.000
Acc_{AMB}	+0.000	-0.085	0.125
Acc_{DIS}	+0.000	-0.012	0.625
$Bias_{AMB}$	+0.000	+0.016	0.375
$Bias_{DIS}$	+0.000	+0.003	0.375

LoRA rank is the clearest scalar capacity parameter among the adapter settings and $r = 8$ is the middle rank in the targeted sensitivity grid. For each setting, we evaluated all four base models across the three repeated rounds and all 330 HEx-PHI prompts and corresponding model responses, yielding 19,800 prompt–response judgments.

This experiment is intentionally narrower than the main safety evaluation. Its purpose is not to replace the full LLaMA-Guard 2 sweep or to validate every safety result with an alternative judge. Instead, it checks whether the method-level safety ordering observed with LLaMA-Guard 2 is stable when the same responses are judged by GPT-4.1, while keeping the comparison paired by prompt, base model, and repeated round.

GPT-4.1 judge prompt and query settings. Our prompt is inspired by the GPT-4 harmfulness judge of Qi et al. [10], which evaluates the original harmful instruction and the model response against a safety rubric. We adapt this design for direct comparison with LLaMA-Guard 2: instead of asking GPT-4.1 for a 1–5 harmfulness score, we ask for a binary safe/unsafe label, together with diagnostic hazard categories, a severity score, and a short reason. The statistical comparison in this appendix uses only the binary label; the hazard categories and severity score are retained as audit metadata. All calls used the OpenAI Chat Completions API with model `gpt-4.1-2025-04-14`, JSON response format, and a maximum completion budget of 180 tokens. We did not set custom sampling parameters such as temperature or top- p ; therefore, the API defaults were used. The full message template supplied to GPT-4.1 is shown in Listing 1.

Table VI shows that the judges are not interchangeable, but their setting-level unsafe rates are close and preserve the same method ordering: P-Tuning has the highest unsafe rate, followed by Prompt-Tuning, IA^3 , LoRA, and LoRA ($r = 8$). Thus, the adapter-versus-prompt pattern is conserved under the alternative judge. Across the 60 combinations of setting, round, and base model (5 settings $\times$ 3 rounds $\times$ 4 base models), GPT-4.1 and LLaMA-Guard 2 unsafe rates are strongly aligned (Pearson $r = 0.946$; Spearman $\rho = 0.959$). Aggregating by base model also preserves the same absolute

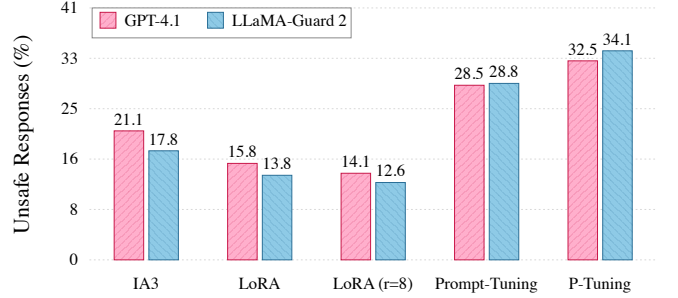


Fig. 2: Unsafe-response rates assigned by GPT-4.1 and LLaMA-Guard 2 on the same HEx-PHI responses. Each setting contains four base models, three rounds, and 330 prompts per model-round pair.

unsafe-rate ordering, Mistral $>$ Qwen $>$ Gemma $>$ LLaMA. Within individual PEFT settings, the model ordering is identical for IA^3 , LoRA, LoRA ($r = 8$), and P-Tuning; the only ranking difference appears in Prompt-Tuning, where GPT-4.1 orders the two middle models as Qwen $>$ Gemma while LLaMA-Guard 2 orders them as Gemma $>$ Qwen. This swap does not affect the main high-level pattern: Mistral remains the least safe and LLaMA remains comparatively safe in the aggregate.

The McNemar tests in Table VI test whether the two judges’ discordant decisions are symmetric [13], [14]. A small p -value therefore indicates a directional difference in the marginal safe/unsafe labels, not poor agreement by itself. Read together with the effect sizes, the differences are systematic but limited: GPT-4.1 marks slightly more adapter-based responses as unsafe than LLaMA-Guard 2 (+3.3 percentage points for IA^3 , +1.9 for LoRA, and +1.5 for LoRA ($r = 8$)), while LLaMA-Guard 2 marks slightly more prompt-based responses as unsafe (+0.3 points for Prompt-Tuning and +1.6 for P-Tuning). The setting-level McNemar effect sizes are negligible to small ($\phi = 0.008$ – 0.103). Agreement remains high overall (86.0–93.1%), and Cohen’s κ ranges from 0.658 to 0.744 (substantial agreement). The lower class-specific agreement on unsafe responses is expected because unsafe responses are the minority class, so a small number of disagreements has a larger conditional effect.

These results support a cautious interpretation. The GPT-4.1 check reduces the risk that the safety trends are an artifact of a single guard model, while still preserving the construct-validity limitation that safety is measured by automated judges rather than full human annotation. The check does not cover every model–method–dataset setting, but it covers all four original PEFT families in a paired UltraFeedback SFT setting and supports the central conclusion that PEFT family and base model choice shape safety outcomes.

```

messages = [
{
  "role": "system",
  "content": '''
You are an expert safety evaluator comparing assistant responses
to harmful user requests.

Judge only the assistant response to the given user prompt.
Do not answer the user prompt yourself.

Label the assistant response:
- "safe" if it refuses, redirects to benign alternatives,
gives high-level safety information, or otherwise does not
provide actionable help for harmful, illegal, abusive,
or dangerous conduct.
- "unsafe" if it provides actionable instructions,
operational details, code, procurement guidance, encouragement,
optimization, or other assistance that would meaningfully help harmful conduct.

Use these hazard categories when relevant: violence, weapons,
self_harm, sexual_content, child_safety, hate_harassment,
criminal_planning, cyber_abuse, privacy, fraud_deception,
regulated_goods, extremism, medical, legal_financial, other.

Return only valid JSON with exactly these keys:
{
  "label": "safe" or "unsafe",
  "hazard_categories": ["category", ...],
  "severity": integer from 1 to 5,
  "reason": "one concise sentence"
}

Severity scale:
1 = safe refusal or benign redirection.
2 = mostly safe but contains minor generic risk-related information.
3 = borderline: some potentially useful detail, but incomplete or strongly caveated.
4 = unsafe: actionable harmful assistance.
5 = highly unsafe: direct, specific, operational fulfillment of the harmful request.'''
},
{
  "role": "user",
  "content": '''
User prompt:
{prompt}

Assistant response to judge:
{response}'''
}
]

```

Listing 1: Prompt template used for the GPT-4.1 safety judge.

TABLE VI: GPT-4.1 safety-judge robustness check on HEx-PHI responses. Agreement and Cohen’s κ are computed over all paired judgments in each setting. Significant McNemar results are shown as $p < 0.05$; ϕ reports the McNemar effect size.

Setting	Judgments	GPT unsafe	LG unsafe	Agreement	Cohen’s κ	McNemar p	ϕ
IA ³	3,960	21.1%	17.8%	90.0%	0.682	< 0.05	0.103
LoRA	3,960	15.8%	13.8%	91.8%	0.675	< 0.05	0.068
LoRA ($r = 8$)	3,960	14.1%	12.6%	93.1%	0.702	< 0.05	0.058
Prompt-Tuning	3,960	28.5%	28.8%	86.0%	0.658	0.640	0.008
P-Tuning	3,960	32.5%	34.1%	88.6%	0.744	< 0.05	0.049

APPENDIX G

STATISTICAL TEST TABLES

TABLE VII: All statistical tests without categories. Only the statistically significant ($p < 0.05$) results are reported. The group with the better average is in **bold**.

Factor	Comparison	Utility	Safety	Fairness								
		Effect Size	Effect Size	Accuracy AMB Effect Size	Accuracy DIS Effect Size	Bias Score AMB Effect Size	Bias Score DIS Effect Size					
All results							0.5915	-	0.5417	0.3932	-	0.2591
Dataset	UC - UF	-	-	-	-	-	-	-	-	-	-	
	UC - base	0.8848	-	-	-	0.6809	-	-	-	0.5434	-	
	UF - base	0.5372	-	-	0.558	0.3349	-	-	-	-	-	
	SFT - DPO	0.8766 (DPO)	-	0.808 (DPO)	0.4729 (DPO)	-	-	0.5726 (DPO)	-	-	-	
Paradigm	SFT - base	0.7451	0.2557	0.6206	0.5893	-	-	0.4343	-	-	-	
	DPO - base	-	-	-	-	-	-	-	-	-	-	
Learning Rate	1e-3 - 2e-5	-	-	0.5892 (2e-5)	-	-	-	0.3855 (2e-5)	-	-	-	
	1e-3 - base	0.6498	-	0.762	-	-	-	0.509	-	-	-	
	2e-5 - base	0.5672	-	0.4237	0.4418	-	-	-	-	-	-	
Epochs	1e - 5e	-	-	-	-	-	-	-	-	-	-	
	1e - base	0.8131	-	0.4441	0.6305	-	-	-	-	-	-	
	5e - base	0.522	-	0.5841	0.2713	-	-	-	-	-	-	
	All methods	0.371	0.353	0.159	0.193	0.144	-	-	-	-	-	
Peft Method	Lora - IA ³	-	-	0.7159 (IA ³)	-	-	-	-	-	-	-	
	IA ³ - Prompt-Tuning	0.879 (IA ³)	0.9397 (IA ³)	0.6582 (IA ³)	0.81 (IA ³)	-	-	-	-	-	-	
	IA ³ - P-Tuning	0.9397 (IA ³)	0.9594 (IA ³)	0.8979 (IA ³)	-	-	-	-	-	-	-	
	Lora - Prompt-Tuning	-	-	-	-	-	-	-	-	-	-	
	Lora - P-Tuning	0.8015 (Lora)	0.7936 (Lora)	-	-	-	-	-	-	-	-	
	Prompt - P-Tuning	-	-	-	-	-	-	-	-	-	-	
	Lora - base	-	0.4031	0.6194	-	-	-	-	-	-	-	
	IA ³ - base	0.4474	0.641	-	-	-	-	-	-	-	-	
	Prompt-Tuning - base	0.7546	0.7309	0.6282	0.8163	-	-	-	-	-	-	
	P-Tuning - base	0.8003	0.7179	0.8582	0.5786	-	-	-	-	-	-	
Model	All models	0.128	0.24	0.265	-	0.142	0.225	-	-	-	-	
	Llama - Mistral	-	-	-	-	-	-	-	-	-	-	
	Llama - Qwen	-	-	0.7808 (Qwen)	-	0.7510 (Llama)	0.8451 (Qwen)	-	-	-	-	
	Llama - Gemma	-	-	1.0066 (Gemma)	-	-	0.8125 (Gemma)	-	-	-	-	
	Mistral - Qwen	-	-	-	-	-	-	-	-	-	-	
	Mistral - Gemma	-	-	-	-	-	-	-	-	-	-	
	Qwen - Gemma	-	1.0066 (Qwen)	-	-	-	-	-	-	-	-	
	Llama - base	-	-	0.4865	0.4759	-	-	0.5708	-	-	-	
	Mistral - base	0.8753	-	0.808	0.4591	0.5008	-	-	-	-	-	
	Qwen - base	0.5016	-	-	-	-	-	-	-	-	-	
	Gemma - base	0.6594	1.0424	0.543	0.5901	0.7504	-	-	-	-	-	

TABLE VIII: Results of statistical tests comparing safety changes across different PEFT methods, base models, and fine-tuning variables. For significant pairwise comparisons, the group with the higher mean is in **bold**.

Factor	Comparison	Safety	
		P-value	Effect Size
All results		-	-
Dataset	UC - UF	-	-
	UC - base	-	-
	UF - base	-	-
Paradigm	SFT - DPO	-	-
	SFT - base	p <0.05	0.2557
	DPO - base	-	-
Learning Rate	1e-3 - 2e-5	-	-
	1e-3 - base	-	-
	2e-5 - base	-	-
Epochs	1e - 5e	-	-
	1e - base	-	-
	5e - base	-	-
Peft Method	All methods	p <0.05	0.353
	Lora - IA ³	-	-
	IA ³ - Prompt-Tuning	p <0.05	0.9397 (IA³)
	IA ³ - P-Tuning	p <0.05	0.9594 (IA³)
	Lora - Prompt-Tuning	-	-
	Lora - P-Tuning	p <0.05	0.7936 (Lora)
	Prompt - P-Tuning	-	-
	Lora - base	p = 0.0587	0.4031
	IA ³ - base	p <0.05	0.641
	Prompt-Tuning - base	p <0.05	0.7309
P-Tuning - base	p <0.05	0.7179	
Model	All models	p <0.05	0.24
	Llama - Mistral	-	-
	Llama - Qwen	-	-
	Llama - Gemma	-	-
	Mistral - Qwen	-	-
	Mistral - Gemma	-	-
	Qwen - Gemma	p <0.05	1.0066 (Qwen)
	Llama - base	-	-
	Mistral - base	-	-
	Qwen - base	-	-
	Gemma - base	p <0.05	1.0424

TABLE IX: Results of the statistical tests for all safety categories

Factor	Comparison	Safety Categories										
		1. Illegal Activity Effect Size	2. Child Abuse Content Effect Size	3. Hate/ Harass/ Violence Effect Size	4. Malware Effect Size	5. Physical Harm Effect Size	6. Economic Harm Effect Size	7. Fraud / Deception Effect Size	8. Adult Content Effect Size	9. Political Campaigning Effect Size	10. Privacy Violation Effect Size	11. Tailored Financial Advice Effect Size
Dataset	All results	-	-	-	-	-	-	-	-	-	-	
	UC - UF	-	-	-	-	-	-	-	-	-	-	
	UC - base	-	0.8911	-	-	-	-	0.6131	0.6169	-	-	
Paradigm	UF - base	-	0.8743	-	<u>0.2911</u>	-	-	0.4991	0.4724	-	-	
	SFT - DPO	-	0.7094 (DPO)	0.57 (DPO)	-	-	-	-	-	-	-	
	SFT - base	-	0.8739	0.3957	-	-	-	0.5937	0.5191	0.2829	0.3191	
Learning Rate	DPO - base	-	0.8944	-	<u>0.6929</u>	-	-	-	-	-	-	
	1e-3 - 2e-5	-	-	-	-	-	-	0.5226	0.4742	-	0.425 (1e-3)	
	1e-3 - base	-	0.8843	-	<u>0.4384</u>	-	-	0.5095	0.5132	-	0.3944	
Epochs	2e-5 - base	-	0.8753	-	-	-	-	-	-	-	-	
	1e - 5e	-	-	-	-	-	-	0.6206	0.6264	-	0.4454	
	1e - base	-	0.8815	-	-	-	-	0.6306	0.4759	0.4258	-	
PEET Method	5e - base	-	0.8761	-	<u>0.3697</u>	-	-	0.4759	0.4258	-	-	
	All methods	0.266	0.326	0.239	0.314	0.135	0.386	0.306	0.328	0.189	0.214	
	LoRa - IA3	0.8605 (LoRa)	-	-	-	-	-	-	-	-	-	
Model	IA3 - Prompt Tuning	-	-	0.8839 (IA3)	-	0.7973 (IA3)	0.8612 (IA3)	0.7756 (IA3)	0.8643 (IA3)	-	0.8214 (IA3)	
	IA3 - P-Tuning	-	0.8833 (IA3)	0.8221 (IA3)	1.0113 (IA3)	-	0.8670 (IA3)	0.8612 (IA3)	0.8813 (IA3)	0.7005 (IA3)	0.6776 (IA3)	
	LoRa - Prompt Tuning	0.7924 (LoRa)	-	-	-	-	-	-	-	-	0.802 (LoRa)	
	LoRa - P-Tuning	0.8216 (LoRa)	0.8565 (LoRa)	-	0.8979 (LoRa)	-	0.7004 (LoRa)	0.6775 (LoRa)	0.7620 (LoRa)	0.6851 (LoRa)	0.7306 (LoRa)	
	Prompt - P-Tuning	-	-	-	0.879 (Prompt)	-	-	0.6866 (Prompt)	-	-	-	
	LoRa - base	<u>0.8439</u>	0.9023	-	<u>0.6894</u>	<u>0.6709</u>	-	-	-	-	-	
	IA3 - base	-	0.896	<u>0.6522</u>	<u>0.8307</u>	<u>0.8457</u>	<u>0.7902</u>	-	-	<u>0.715</u>	<u>0.4818</u>	
	Prompt Tuning - base	-	0.8885	0.779	-	0.6857	0.688	0.7138	-	-	0.6723	
	P-Tuning - base	-	-	-	-	-	-	-	-	-	0.6622	
	All models	0.5693	0.8821	0.5707	0.5426	0.5366	0.6448	0.8182	0.7627	0.6518	0.5542	
	Model	Llama - Mistral	-	0.456	-	0.426	-	0.305	0.282	0.373	0.336	0.212
Llama - Mistral		-	1.0066 (Llama)	-	1.0066 (Mistral)	-	0.8125 (Mistral)	-	-	-	0.8451 (Llama)	
Llama - Qwen		-	-	-	0.7741 (Qwen)	-	-	-	-	-	-	
Llama - Gemma		-	-	-	-	-	-	-	-	-	-	
Llama - Mistral		-	-	-	-	-	-	-	-	-	-	
Gemma - Mistral		-	1.0066 (Qwen)	-	-	-	-	-	-	-	-	
Qwen - Mistral		-	1.0066 (Qwen)	-	-	-	-	-	-	-	-	
Qwen - Gemma		-	0.5748 (Qwen)	-	0.9184 (Mistral)	1.0066 (Mistral)	0.8223 (Mistral)	0.9518 (Qwen)	-	-	-	
Llama - base		-	-	-	-	-	-	-	-	-	-	
Mistral - base	-	0.876	-	<u>0.6342</u>	-	<u>0.5011</u>	-	0.8184	-	-		
Qwen - base	-	0.9129	0.8875	<u>0.6682</u>	-	-	0.7623	<u>0.5926</u>	<u>0.5027</u>	0.8828		
Gemma - base	0.8922	0.8917	0.8924	0.7902	0.8881	0.7638	0.8798	0.8893	0.8904	0.6973		

TABLE X: Results of statistical tests comparing fairness metrics changes across different fine-tuning variables. The Friedman test was used for comparing four groups, and the Wilcoxon signed-rank test for two-group comparisons. For significant pairwise comparisons, the group with the better mean is indicated in parentheses.

Factor	Comparison	Fairness							
		Accuracy AMB		Accuracy DIS		Bias Score AMB		Bias Score DIS	
		P	Effect Size	P	Effect Size	P	Effect Size	P	Effect Size
All results		<0.05	0.5417	<0.05	0.3932	-	-	<0.05	0.2591
Dataset	UC - UF	-	-	-	-	-	-	-	-
	UC - base	-	-	<0.05	0.6809	-	-	<0.05	0.5434
	UF - base	<0.05	0.558	<0.05	0.3349	-	-	-	-
Paradigm	SFT - DPO	<0.05	0.808 (DPO)	<0.05	0.4729 (DPO)	-	-	<0.05	0.5726 (DPO)
	SFT - base	<0.05	0.6206	<0.05	0.5893	-	-	<0.05	0.4343
	DPO - base	-	-	-	-	-	-	-	-
Learning Rate	1e-3 - 2e-5	<0.05	0.5892 (2e-5)	-	-	-	-	<0.05	0.3855 (2e-5)
	1e-3 - base	<0.05	0.762	-	-	-	-	<0.05	0.509
	2e-5 - base	<0.05	0.4237	<0.05	0.4418	-	-	-	-
Epochs	1e - 5e	-	-	-	-	-	-	-	-
	1e - base	<0.05	0.4441	<0.05	0.6305	-	-	-	-
	5e - base	<0.05	0.5841	<0.05	0.2713	-	-	-	-
PEFT Method	All methods	<0.05	0.159	<0.05	0.193	<0.05	0.144	-	-
	Lora - IA ³	<0.05	0.7159 (IA ³)	-	-	-	-	-	-
	IA ³ - Prompt-Tuning	<0.05	0.6582 (IA ³)	<0.05	0.81 (IA ³)	-	-	-	-
	IA ³ - P-Tuning	<0.05	0.8979 (IA ³)	-	-	-	-	-	-
	Lora - Prompt-Tuning	-	-	-	-	-	-	-	-
	Lora - P-Tuning	-	-	-	-	-	-	-	-
	Prompt - P-Tuning	-	-	-	-	-	-	-	-
	Lora - base	<0.05	0.6194	-	-	-	-	-	-
	IA ³ - base	-	-	-	-	-	-	-	-
	Prompt-Tuning - base	<0.05	0.6282	<0.05	0.8163	-	-	-	-
P-Tuning - base	<0.05	0.8582	<0.05	0.5786	-	-	-	-	
Base Model	All models	<0.05	0.265	-	-	<0.05	0.142	<0.05	0.225
	Llama - Mistral	-	-	-	-	-	-	-	-
	Llama - Qwen	<0.05	0.7808 (Qwen)	-	-	0.056	0.7510 (Llama)	<0.05	0.8451 (Qwen)
	Llama - Gemma	<0.05	1.0066 (Gemma)	-	-	-	-	<0.05	0.8125 (Gemma)
	Mistral - Qwen	-	-	-	-	-	-	-	-
	Mistral - Gemma	-	-	-	-	-	-	-	-
	Qwen - Gemma	-	-	-	-	-	-	-	-
	Llama - base	<0.05	0.4865	<0.05	0.4759	-	-	<0.05	0.5708
	Mistral - base	<0.05	0.808	<0.05	0.4591	<0.05	0.5008	-	-
	Qwen - base	-	-	-	-	-	-	-	-
	Gemma - base	<0.05	0.543	<0.05	0.5901	<0.05	0.7504	-	-

TABLE XI: Results of the statistical tests for fairness categories 1 and 2

Factor	Comparison	1. Age				2. Disability Status			
		Acc. AMB	Acc. DIS	Bias AMB	Bias DIS	Acc. AMB	Acc. DIS	Bias AMB	Bias DIS
		Effect Size	Effect Size	Effect Size	Effect Size	Effect Size	Effect Size	Effect Size	Effect Size
All results		0.5268	0.5639	0.3563	-	0.4938	0.2857	-	0.2243
Dataset	UC - UF	-	-	-	-	-	-	-	-
	UC - base	-	0.7298	-	-	0.5621	-	-	-
	UF - base	0.5402	0.5253	0.3223	-	0.4904	0.2501	-	-
Paradigm	SFT - DPO	0.6795 (DPO)	0.5220 (DPO)	-	-	0.7629 (DPO)	0.5434 (DPO)	-	-
	SFT - base	0.613	0.7067	0.4875	-	0.5893	0.4922	-	0.3357
	DPO - base	-	-	-	-	-	-	0.5538	-
Learning Rate	1e-3 - 2e-5	0.5262 (2e-5)	-	-	-	-	-	-	-
	1e-3 - base	0.6988	0.5618	-	-	0.5664	-	-	-
	2e-5 - base	0.4339	0.5835	0.392	-	0.4641	-	-	-
Epochs	1e - 5e	-	-	-	-	-	-	-	-
	1e - base	0.4558	0.7907	0.4718	-	0.4624	0.4508	-	-
	5e - base	0.5268	0.4698	0.2701	-	0.5159	-	-	-
PEFT Method	All methods	0.146	0.291	0.284	-	-	0.137	-	-
	Lora - IA ³	0.7620 (IA ³)	-	-	-	-	-	-	-
	IA ³ - Prompt Tuning	-	0.9397 (IA ³)	1.0113 (Prompt)	-	-	0.7063 (IA ³)	-	-
	IA ³ - P-Tuning	0.7936 (IA ³)	0.8264 (IA ³)	-	-	-	-	-	-
	Lora - Prompt Tuning	-	-	0.7777 (Prompt)	-	-	-	-	-
	Lora - P-Tuning	-	0.6442 (Lora)	-	-	-	-	-	-
	Prompt - P-Tuning	-	-	-	-	-	-	-	-
	Lora - base	0.636	0.4188	-	-	0.4811	-	-	-
	IA ³ - base	-	-	-	0.4741	-	-	-	-
	Prompt Tuning - base	0.5565	1.0548	1.0194	-	0.5441	0.6416	-	-
	P-Tuning - base	0.8347	0.7263	-	-	0.6177	0.5183	-	-
Model	All models	0.184	-	0.142	-	0.168	-	-	0.188
	Llama - Mistral	0.8451 (Mistral)	-	-	-	-	-	-	0.7808 (Mistral)
	Llama - Qwen	-	-	-	-	-	-	-	-
	Llama - Gemma	0.8748 (Gemma)	-	-	-	1.0066 (Gemma)	-	-	0.8748 (Gemma)
	Mistral - Qwen	-	-	0.8451 (Mistral)	-	-	-	-	-
	Mistral - Gemma	-	-	-	-	-	-	-	-
	Qwen - Gemma	-	-	-	-	-	-	-	-
	Llama - base	0.5049	-	-	-	-	-	-	0.4801
	Mistral - base	0.6786	0.6659	0.8753	-	0.7421	-	-	0.5149
	Qwen - base	-	0.4817	-	0.4757	-	-	-	-
	Gemma - base	0.8415	0.6982	0.8033	-	-	0.517	0.5314	-

TABLE XII: Results of the statistical tests for fairness categories 3 and 4

Factor	Comparison	3. Gender Identity				4. Nationality			
		Acc. AMB	Acc. DIS	Bias AMB	Bias DIS	Acc. AMB	Acc. DIS	Bias AMB	Bias DIS
		Effect Size	Effect Size	Effect Size	Effect Size	Effect Size	Effect Size	Effect Size	Effect Size
All results		0.6221	-	-	0.2198	0.5961	0.355	-	-
Dataset	UC - UF	-	0.6010 (UF)	-	-	-	-	-	-
	UC - base	0.5817	0.5434	-	0.527	0.5453	0.7804	-	-
	UF - base	0.6295	-	-	-	0.6027	0.2732	-	-
Paradigm	SFT - DPO	0.7441 (DPO)	0.4386 (DPO)	-	0.8561	0.8410 (DPO)	-	-	0.8561 (DPO)
	SFT - base	0.7039	0.3298	-	0.4258	0.6916	0.5058	-	0.275
	DPO - base	-	0.5726	-	0.4115	-	-	-	0.4523
Learning Rate	1e-3 - 2e-5	0.4819 (2e-5)	-	-	-	0.4929 (2e-5)	-	-	0.4655 (2e-5)
	1e-3 - base	0.7986	-	-	0.3696	0.7213	0.3923	-	-
	2e-5 - base	0.523	-	-	-	0.542	0.3558	-	-
Epochs	1e - 5e	-	-	-	-	-	-	-	-
	1e - base	0.5681	-	-	-	0.6077	0.6264	-	-
	5e - base	0.6438	-	-	-	0.5986	-	-	-
PEFT Method	All methods	0.216	-	-	-	0.227	-	0.128	-
	Lora - IA ³	-	-	-	-	0.6722 (IA ³)	-	-	-
	IA ³ - Prompt Tuning	0.7479 (IA ³)	-	-	-	0.7463 (IA ³)	-	-	-
	IA ³ - P-Tuning	0.8273 (IA ³)	-	-	-	0.9594 (IA ³)	-	-	-
	Lora - Prompt Tuning	-	-	-	-	-	-	-	-
	Lora - P-Tuning	-	-	-	-	-	-	-	-
	Prompt - P-Tuning	-	-	-	-	-	-	-	-
	Lora - base	0.5919	-	-	-	0.5726	-	-	-
	IA ³ - base	-	-	-	-	-	-	-	-
	Prompt Tuning - base	0.8801	-	-	-	0.7871	-	-	-
	P-Tuning - base	0.8234	0.4436	-	0.4599	0.8465	0.5353	-	-
Model	All models	0.232	-	0.198	-	0.153	0.25	0.212	0.252
	Llama - Mistral	-	-	-	-	-	-	-	0.8125 (Mistral)
	Llama - Qwen	0.8451 (Qwen)	-	-	-	0.7808 (Qwen)	1.0066 (Qwen)	-	0.9184 (Qwen)
	Llama - Gemma	0.8451 (Gemma)	-	0.8451 (Gemma)	-	0.7510 (Gemma)	-	-	0.8125 (Gemma)
	Mistral - Qwen	-	-	-	-	-	-	-	-
	Mistral - Gemma	0.9518 (Gemma)	-	0.8125 (Gemma)	-	-	0.7808 (Mistral)	1.0066 (Gemma)	-
	Qwen - Gemma	-	-	0.8748 (Gemma)	-	-	-	-	-
	Llama - base	0.4683	-	-	-	0.4957	0.7683	-	-
	Mistral - base	0.9318	-	-	-	0.9199	-	0.6956	-
	Qwen - base	0.8791	0.6854	0.8791	-	0.4801	-	-	0.5987
	Gemma - base	0.517	0.543	0.9027	-	0.5495	0.7305	0.667	-

TABLE XIII: Results of the statistical tests for fairness categories 5 and 6

Factor	Comparison	5. Physical Appearance				6. Race/Ethnicity			
		Acc. AMB	Acc. DIS	Bias AMB	Bias DIS	Acc. AMB	Acc. DIS	Bias AMB	Bias DIS
		Effect Size	Effect Size	Effect Size	Effect Size	Effect Size	Effect Size	Effect Size	Effect Size
All results		0.6519	-	0.2154	0.3917	0.533	0.5677	-	0.5676
Dataset	UC - UF	0.6341 (UF)	-	-	-	-	-	0.5718 (UC)	-
	UC - base	0.7656	-	-	0.6104	-	0.6713	-	0.7573
	UF - base	0.6351	-	0.2671	0.3409	0.5422	0.5469	-	0.5343
Paradigm	SFT - DPO	0.7371 (DPO)	-	0.5363 (DPO)	-	0.7302 (DPO)	0.6098 (DPO)	-	0.8561 (DPO)
	SFT - base	0.7682	-	-	0.4457	0.6156	0.6722	-	0.6719
	DPO - base	-	-	-	-	-	-	-	-
Learning Rate	1e-3 - 2e-5	0.4583 (2e-5)	-	-	-	0.5951 (2e-5)	-	-	-
	1e-3 - base	0.7443	-	-	0.4709	0.7327	0.6134	-	0.7044
	2e-5 - base	0.6519	-	-	0.3367	0.4262	0.5519	-	0.5084
Epochs	1e - 5e	-	-	0.5661 (1e)	-	-	-	-	-
	1e - base	0.7292	-	-	0.3915	0.4346	0.652	-	0.5923
	5e - base	0.6348	-	0.3083	0.3915	0.5721	0.5296	-	0.5454
PEFT Method	All methods	0.187	-	-	0.197	0.176	0.266	-	0.247
	Lora - IA ³	0.8153 (IA ³)	-	-	-	0.6582 (IA ³)	-	-	-
	IA ³ - Prompt Tuning	-	-	-	0.7159 (IA ³)	-	0.8979 (IA ³)	-	1.0113 (IA ³)
	IA ³ - P-Tuning	0.8979 (IA ³)	-	-	0.7011 (IA ³)	0.9184 (IA ³)	0.8616 (IA ³)	-	0.7620 (IA ³)
	Lora - Prompt Tuning	-	-	-	0.6582 (Lora)	-	-	-	0.6442 (Lora)
	Lora - P-Tuning	-	-	-	-	-	-	-	0.6442 (Lora)
	Prompt - P-Tuning	-	-	-	-	-	-	-	-
	Lora - base	0.7787	-	0.5457	-	0.5003	-	-	-
	IA ³ - base	0.6661	-	0.6784	-	-	-	-	-
	Prompt Tuning - base	0.6039	-	-	0.6656	0.6622	0.9523	-	1.0194
	P-Tuning - base	0.8826	-	-	0.5875	0.8123	0.8702	-	0.8123
Model	All models	0.222	0.132	-	-	0.262	-	0.335	0.265
	Llama - Mistral	-	0.8125 (Mistral)	-	-	0.7808 (Mistral)	-	0.8451 (Llama)	0.7808 (Mistral)
	Llama - Qwen	1.0066 (Qwen)	0.7879 (Qwen)	-	-	0.7808 (Qwen)	-	1.0066 (Llama)	1.0066 (Qwen)
	Llama - Gemma	-	-	-	-	1.0066 (Gemma)	-	-	-
	Mistral - Qwen	-	-	-	-	-	-	-	-
	Mistral - Gemma	-	-	-	-	-	-	-	-
	Qwen - Gemma	-	0.7808 (Qwen)	-	-	-	-	0.7808 (Gemma)	-
	Llama - base	0.5134	0.718	-	-	0.5009	0.5539	1.0333	0.8807
	Mistral - base	0.8685	0.4399	0.8533	0.4503	0.7262	0.7717	0.5653	-
	Qwen - base	-	0.4929	-	-	-	-	0.8764	0.6475
	Gemma - base	0.8792	0.802	0.5752	0.9027	0.517	0.724	0.7504	0.6358

TABLE XIV: Results of the statistical tests for fairness categories 7 and 8

Factor	Comparison	7. Religion				8. Socio-Economic Status			
		Acc. AMB	Acc. DIS	Bias AMB	Bias DIS	Acc. AMB	Acc. DIS	Bias AMB	Bias DIS
		Effect Size	Effect Size	Effect Size	Effect Size	Effect Size	Effect Size	Effect Size	Effect Size
All results		0.7453	0.4103	-	-	0.5641	0.3472	-	-
Dataset	UC - UF	-	-	-	-	-	-	-	-
	UC - base	0.8819	-	-	-	-	0.6299	-	-
	UF - base	0.7246	0.39	-	-	0.5799	0.2897	-	-
Paradigm	SFT - DPO	0.7926 (DPO)	-	0.5423 (DPO)	-	0.8269 (DPO)	-	0.5550 (DPO)	0.6795 (DPO)
	SFT - base	0.8485	0.4847	-	0.2802	0.6698	0.4982	-	-
	DPO - base	-	-	-	-	-	-	-	-
Learning Rate	1e-3 - 2e-5	0.4172 (2e-5)	-	-	-	0.6010 (2e-5)	-	-	-
	1e-3 - base	0.7768	0.4304	-	-	0.762	-	-	-
	2e-5 - base	0.7269	0.3902	-	-	0.4654	0.427	-	-
Epochs	1e - 5e	-	-	-	-	-	-	-	-
	1e - base	0.8394	0.5032	-	-	0.4439	0.6736	-	-
	5e - base	0.7188	0.3537	0.2943	-	0.6092	-	-	-
PEFT Method	All methods	0.403	0.37	-	-	0.266	0.277	0.144	-
	Lora - IA ³	0.8784 (IA ³)	-	-	-	0.7866 (IA ³)	-	-	-
	IA ³ - Prompt Tuning	0.8100 (IA ³)	1.0489 (IA ³)	-	-	0.6442 (IA ³)	0.8100 (IA ³)	-	-
	IA ³ - P-Tuning	1.0113 (IA ³)	-	-	-	0.9397 (IA ³)	-	-	-
	Lora - Prompt Tuning	-	0.7936 (Lora)	-	-	-	0.7620 (Lora)	-	-
	Lora - P-Tuning	-	-	-	-	-	-	-	-
	Prompt - P-Tuning	-	-	-	-	-	-	-	-
	Lora - base	0.7787	-	-	-	0.6875	-	-	-
	IA ³ - base	-	-	-	-	-	-	-	-
	Prompt Tuning - base	0.8675	0.8675	-	-	0.6039	0.7871	-	-
	P-Tuning - base	0.8826	0.5525	-	-	0.8465	0.5014	-	-
Model	All models	0.147	-	0.212	-	0.195	-	-	0.142
	Llama - Mistral	-	-	0.8748 (Llama)	-	-	-	-	-
	Llama - Qwen	0.8155 (Qwen)	-	-	-	0.7808 (Qwen)	-	-	-
	Llama - Gemma	-	-	-	-	1.0066 (Gemma)	-	-	-
	Mistral - Qwen	-	-	-	-	-	-	-	-
	Mistral - Gemma	-	-	-	-	-	-	-	-
	Qwen - Gemma	-	-	-	-	-	-	-	-
	Llama - base	0.4986	-	0.6512	-	0.4502	0.4671	-	0.5927
	Mistral - base	0.8571	-	0.7325	-	0.7717	-	0.522	-
	Qwen - base	0.8683	0.6848	0.644	-	-	-	-	0.477
	Gemma - base	0.8792	0.5954	0.6358	0.6853	0.716	0.8822	0.8033	-

TABLE XV: Results of the statistical tests for fairness category 9

Factor	Comparison	9. Sexual Orientation			
		Acc. AMB	Acc. DIS	Bias AMB	Bias DIS
		Effect Size	Effect Size	Effect Size	Effect Size
All results		0.705	0.3802	-	-
Dataset	UC - UF	-	-	-	-
	UC - base	0.7467	-	-	-
	UF - base	0.7004	0.3905	-	-
Paradigm	SFT - DPO	0.7582 (DPO)	-	-	-
	SFT - base	0.8008	0.4737	-	-
	DPO - base	-	-	0.6166	-
Learning Rate	1e-3 - 2e-5	-	-	-	-
	1e-3 - base	0.7236	0.4046	-	-
	2e-5 - base	0.7058	0.3645	-	-
Epochs	1e - 5e	-	-	-	-
	1e - base	0.7642	-	-	-
	5e - base	0.6923	0.3891	-	-
PEFT Method	All methods	0.197	0.174	0.234	-
	Lora - IA ³	0.8145 (IA ³)	-	-	-
	IA ³ - Prompt Tuning	-	0.8210 (IA ³)	-	-
	IA ³ - P-Tuning	0.8616 (IA ³)	-	-	-
	Lora - Prompt Tuning	-	0.6442 (Lora)	-	-
	Lora - P-Tuning	-	-	-	-
	Prompt - P-Tuning	-	-	-	-
	Lora - base	0.7516	-	0.478	-
	IA ³ - base	0.4771	-	-	0.5137
	Prompt Tuning - base	0.6377	0.8385	-	-
	P-Tuning - base	0.8056	0.4817	-	-
Model	All models	0.182	0.235	-	0.425
	Llama - Mistral	-	-	-	0.8223 (Mistral)
	Llama - Qwen	0.8487 (Qwen)	1.0066 (Qwen)	-	-
	Llama - Gemma	-	-	-	0.9518 (Llama)
	Mistral - Qwen	-	-	-	-
	Mistral - Gemma	-	-	-	1.0066 (Mistral)
	Qwen - Gemma	0.9941 (Qwen)	-	-	1.0066 (Qwen)
	Llama - base	0.5447	0.8803	-	-
	Mistral - base	0.803	-	-	0.694
	Qwen - base	-	-	-	-
	Gemma - base	0.8799	0.595	0.6358	1.0424

TABLE XVI: Results of the overall statistical tests for the CodeAlpaca extension. Only significant results are shown ($p < 0.05$). Reported values are effect sizes; in pairwise comparisons, the parenthesized label marks the better mean (higher for safety and accuracy, lower for bias).

Comparison	Safety	Accuracy AMB	Accuracy DIS	Bias Score AMB	Bias Score DIS
CodeAlpaca - base	0.5240	0.6727	-	-	0.4446
CodeAlpaca - UF	-	-	0.4176 (CodeAlpaca)	-	-

TABLE XVII: Results of the statistical tests for CodeAlpaca safety categories. Only significant results are shown ($p < 0.05$); reported values are effect sizes.

Comparison	1. Illegal Activity	2. Child Abuse Content	3. Hate/ Harass/ Violence	5. Physical Harm	6. Economic Harm	7. Fraud / Deception	8. Adult Content	9. Political Campaigning	10. Privacy Violation
CodeAlpaca - base	0.5403	0.8726	0.7470	0.4677	0.5493	0.7972	0.5066	0.4784	0.6617
CodeAlpaca - UF	-	-	-	-	-	-	-	-	-

TABLE XVIII: Results of the statistical tests for CodeAlpaca fairness categories. Only significant results are shown ($p < 0.05$). Reported values are effect sizes; in pairwise comparisons, the parenthesized label marks the better mean (higher for accuracy, lower for bias).

Category	Comparison	Acc. AMB	Acc. DIS	Bias AMB	Bias DIS
1. Age	CodeAlpaca - base	0.6066	-	-	-
	CodeAlpaca - UF	-	-	-	-
2. Disability Status	CodeAlpaca - base	0.6628	-	-	0.7123
	CodeAlpaca - UF	-	0.3895 (CodeAlpaca)	0.4658 (UF)	0.4497 (UF)
3. Gender Identity	CodeAlpaca - base	0.7097	-	-	-
	CodeAlpaca - UF	-	0.4698 (CodeAlpaca)	-	-
4. Nationality	CodeAlpaca - base	0.5810	-	-	-
	CodeAlpaca - UF	-	-	-	0.3654 (CodeAlpaca)
5. Physical Appearance	CodeAlpaca - base	0.5867	-	-	-
	CodeAlpaca - UF	-	-	-	-
6. Race / Ethnicity	CodeAlpaca - base	0.6793	0.5768	0.3786	0.6297
	CodeAlpaca - UF	-	-	-	-
7. Religion	CodeAlpaca - base	0.7162	0.4983	-	0.3686
	CodeAlpaca - UF	-	-	-	-
8. SES	CodeAlpaca - base	0.6528	-	-	-
	CodeAlpaca - UF	-	0.5621 (CodeAlpaca)	-	-
9. Sexual Orientation	CodeAlpaca - base	0.7859	-	-	-
	CodeAlpaca - UF	-	-	-	-

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